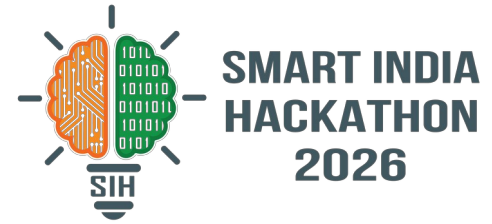




QFLEET



PROBLEM STATEMENT - Quantum-inspired fuel consumption prediction and green fleet optimization

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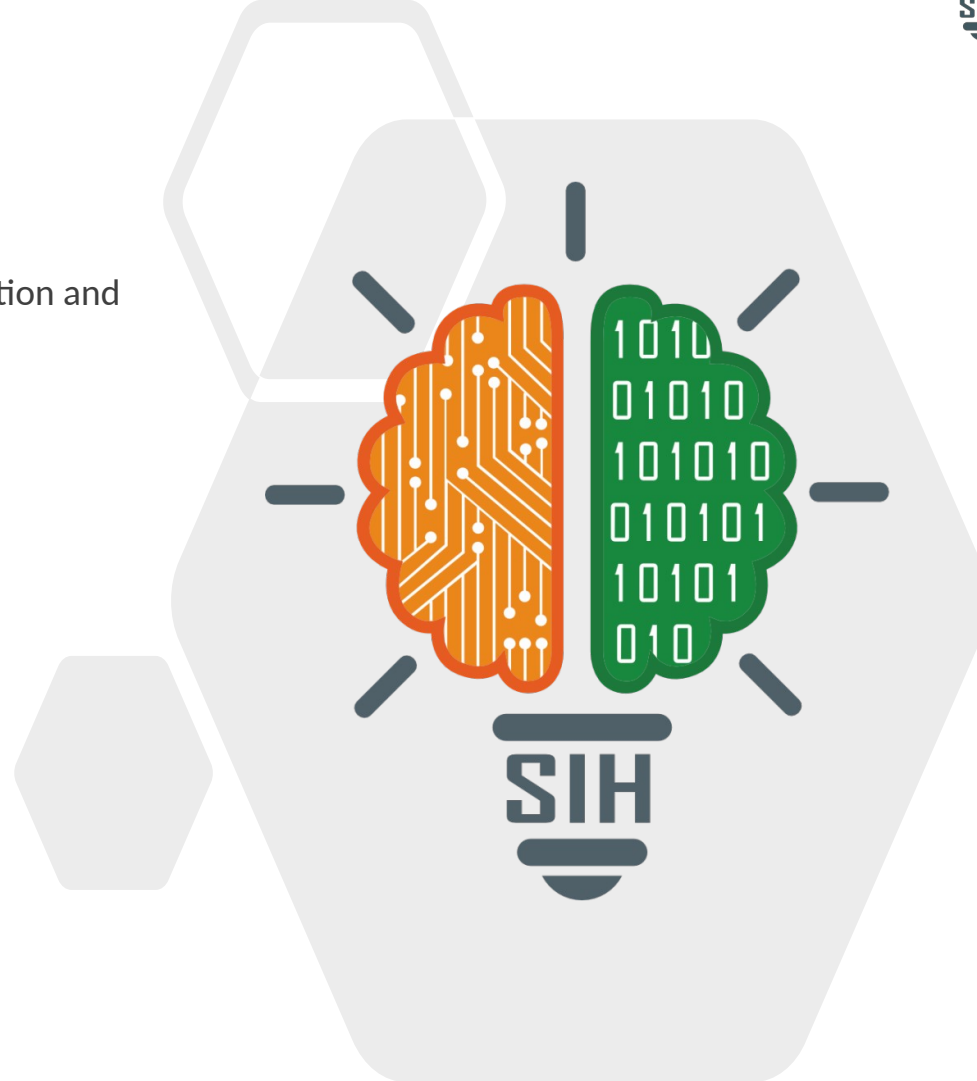
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❖ WHY THIS PROBLEM?

THE PROBLEM

- Cost, well-to-wake GHG and schedule reliability are traded off ad hoc by dispatchers — not jointly optimized.
- Optimizing any one variable alone (just speed, or just fuel) leaves cost or emissions gains on the table.

WHY IT MATTERS

- IMO CII, EU ETS and FuelEU are now mandatory reporting lines for every vessel.
- Carbon is priced directly into operations at ~€85 per tonne — inefficiency is now a compliance cost.

WHY IT IS HARD

- Fuel per hour $\propto V^3$ — the speed-fuel relationship is non-convex.
- $\approx 19\%$ fuel and CO_2 saved by a modest speed reduction (slow steaming).
- But slower ships need more vessels to hold the same schedule.
- Speed, fuel choice and fleet size interact — the trade-off is non-trivial.

❖ EXISTING SOLUTIONS

CLASSICAL MILP

- Linear / mixed-integer formulations used for fleet planning at scale.
- **Gap:** cannot represent the non-convex V^3 speed-fuel relationship — they over-simplify the physics or time out.

CLASSICAL ML BASELINES

- XGBoost / LightGBM predict fuel consumption well in isolation.
- **Gap:** they only forecast — they don't jointly decide speed, fuel and deployment.

SINGLE-OBJECTIVE HEURISTICS

- GA / PSO variants used in practice optimize one objective (usually cost).
- **Gap:** emissions are bolted on afterward — no true Pareto trade-off between cost, GHG and reliability.

❖ PROPOSED SOLUTION — QFLEET

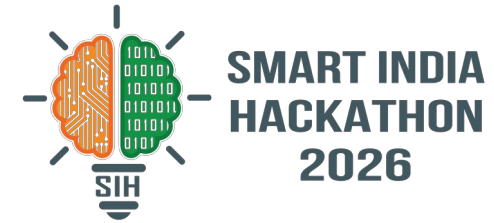
1. **ASSIGN** — QIEA assigns vessels, routes and fuels — the discrete decisions.
2. **TUNE** — QPSO tunes sailing speed (continuous) inside the same optimization loop.
3. **EVOLVE** — Q-NSGA-II evolves a full Pareto front across cost, well-to-wake GHG and schedule reliability — operators pick the trade-off they want.
4. **COMPLY** — Every Pareto point ships with its CII rating, EEXI pass/fail and FuelEU gap already computed.

GROUNDING IN REAL PHYSICS — NOT A BLACK BOX

- Built on the admiralty V^3 law and resistance modelling — hull fouling, draft and sea state.
- Full tank-to-wake → well-to-wake emissions accounting for every candidate solution.



Technologies chosen for hackathon feasibility



❖ TECH STACK

Optimization Core	pymoo • pyqubo • dwave-neal	Hardware-ready QUBO; simulated annealing keeps demo runtime deterministic
Quantum-Inspired ML	PennyLane • Qiskit-Aer	Small statevector circuits (≤ 10 qubits) as feature maps — fast to simulate
Classical ML Baseline	XGBoost • LightGBM • SHAP	The benchmark QFleet has to beat, with SHAP plots for interpretability
Data Pipeline	Polars • GeoPandas • Xarray	AIS + Copernicus weather at scale, stored in Parquet — not CSV
Backend	FastAPI • Celery • Redis	Optimization runs (2–5 min) go to an async job queue, never a blocking request
Reporting & Infra	WeasyPrint • Docker Compose	Auto-generated CII / EEXI / FuelEU compliance PDFs; one command boots the stack

Runs entirely on a laptop — quantum-ready, classically executed

❖ SCALABILITY & SUSTAINABILITY

SCALABILITY

- Contract-first architecture — teams scale independently against one shared JSON schema.
- Async job queue (Celery + Redis) — ready for larger fleets and more routes.
- QUBO is hardware-ready — real quantum annealers are a backend swap, not a redesign.

SUSTAINABILITY

- Multi-fuel scenario engine (VLSFO / MGO / LNG / methanol) tests real transition pathways.
- Full well-to-wake accounting — including CH₄ and N₂O slip — no hidden upstream emissions.
- Automatic CII / EEXI / FuelEU compliance — every recommendation is audit-ready.

IMPACT METRICS

- Fuel cost per voyage
- Well-to-wake CO₂e per route
- CII rating distribution across the fleet
- Schedule reliability (%)
- EU ETS + FuelEU penalty exposure

❖ FUTURE SCOPE & LIMITATIONS

FUTURE SCOPE

- Full React + Tailwind dashboard beyond the Streamlit MVP.
- Extend fuel modelling to hydrogen and ammonia, alongside LNG and methanol.
- Execute the QUBO formulation on real quantum annealing hardware.
- Live AIS and weather feeds for continuous, real-time re-optimization instead of batch runs.

CURRENT LIMITATIONS

- Runs on classical simulated annealing — the QUBO is hardware-ready but untested on an actual annealer.
- Hackathon scope prioritizes LNG + methanol over hydrogen / ammonia, with fewer benchmark seeds than a production study.
- Synthetic training data is physics-calibrated and MRV-aggregate validated, but not yet validated at full fleet scale.
- Demo results are pre-computed for reliability; only smaller instances are optimized live.